Requirements Specification

Business, functional, non-functional, technical, security and AI requirements

Document	03 - Requirements
Project	AI Customer Support Platform (fictional company: Kestrel Commerce Ltd.)
Prepared by	Project Manager (portfolio author) - self-directed simulation
Status	Baseline v1.0 - simulated approval 16 Jan 2026 (M2)
Project window	5 Jan - 13 Mar 2026 (simulated project window)

SIMULATED PROJECT - Self-directed portfolio case study. Not professional experience, not client work, not employment. All people, figures and outcomes are fictional.

Requirements are numbered, testable and traceable to user stories (05_User_Stories.xlsx). Priority uses MoSCoW. Numbers marked as targets are Proposed Project Targets for this simulation, not measured results.

Business Requirements

ID	Requirement	Priority	Source	Verification	Realised by
BR-01	Resolve routine customer questions through self-service so fewer tickets need an agent. Proposed Project Target: at least 35% of conversations resolved without human handover within 8 weeks of pilot go-live.	Must	Sponsor, Support Manager	Analytics (post-launch)	FR-07, FR-08, FR-19
BR-02	Give customers an answer or a human handover in seconds instead of hours. Proposed Project Target: AI answer in 8 seconds or less (p95); first human response to escalations within 2 business hours.	Must	Sponsor, Customer Success	Load test + analytics	FR-11, FR-14, FR-15, NFR-01
BR-03	Let customers reach a human agent whenever the AI cannot or should not answer, without repeating themselves.	Must	Support Manager, Agents	UAT	FR-14, FR-15, FR-16, FR-17
BR-04	Let support staff maintain knowledge-base content without developer involvement.	Must	KB Admin, Support Manager	UAT	FR-04, FR-05, FR-06
BR-05	Give the Support Manager visibility of volume, deflection, escalation, feedback and content gaps.	Must	Support Manager, Customer Success	UAT	FR-18, FR-19
BR-06	Handle customer data in line with the company privacy policy (data minimisation, retention, deletion, controlled access).	Must	Security & Privacy Officer	Test + review	FR-10, FR-20, FR-21, FR-22, SR-06
BR-07	Release a pilot-ready MVP within 10 weeks and within the 5,500 USD non-labour budget (planning assumption).	Must	Sponsor, Finance	Milestone + budget tracking	NFR-02, TR-01, TR-02
BR-08	Agents and admins can adopt the platform quickly. Proposed Project Target: core tasks learnt within a 60-minute training session.	Should	Support Manager	Usability test + training	NFR-04

Functional Requirements

ID	Requirement	Priority	Source	Verification	Stories
FR-01	Staff (Agent, Manager, Admin) can sign in with work email and password and sign out; sessions expire after 30 minutes of inactivity; accounts lock for 15 minutes after 5 failed attempts.	Must	Security Officer	Test	US-04
FR-02	The system enforces role-based access for Customer, Agent, Manager and Admin according to the permission matrix; customers can only access their own data.	Must	Security Officer	Test	US-05
FR-03	Customers sign in with their email address and a single-use one-time code valid for 10 minutes.	Must	Product Manager	Test	US-06
FR-04	Admins can create, edit, publish and unpublish KB articles with title, category, body, tags and review date.	Must	KB Admin	Test + demo	US-09
FR-05	Admins can bulk-import articles from CSV or Markdown; invalid rows are rejected with a reason and imported articles default to Draft.	Should	KB Admin	Test	US-10
FR-06	Published and unpublished articles are reflected in the AI search index within 5 minutes; the full index can be rebuilt in under 30 minutes.	Must	Product Manager	Test	US-12, US-09
FR-07	The AI assistant answers using only retrieved published KB content and displays at least one source link per answer.	Must	Support Manager	Golden-set evaluation	US-13
FR-08	When retrieval confidence is below the configured threshold, the assistant states it cannot answer and offers escalation instead of generating an answer.	Must	Support Manager	Golden-set evaluation	US-14
FR-09	The assistant declines out-of-scope requests (legal advice, refund approval, account-specific actions) and offers escalation.	Must	Support Manager, Legal	Adversarial test set	US-14
FR-10	Emails, phone numbers and card-number patterns are redacted from customer text before it is sent to the LLM provider.	Must	Security Officer	Seeded PII test	US-25
FR-11	Customers can ask questions in a web chat window and receive an answer with sources; AI answers are labelled as AI-generated.	Must	Customer persona	UAT	US-16
FR-12	Customers can rate each AI answer (thumbs up/down) and add an optional comment.	Should	Customer Success	Test	US-17
FR-13	Conversations are stored and customers can view their own conversations from the last 30 days.	Must	Product Manager	Test	US-18
FR-14	Customers can request a human at any time; escalation is also offered after a fallback answer or two consecutive thumbs-down ratings.	Must	Support Manager	UAT	US-19
FR-15	Escalated conversations appear in an agent queue with full transcript, category and waiting time; one agent can claim an item at a time.	Must	Agents	UAT	US-20
FR-16	Agents can reply to the customer and mark an escalation resolved with a mandatory resolution category.	Must	Agents	UAT	US-21
FR-17	Customers see agent replies in the open chat and receive an email notification when away.	Must	Agents, Customers	UAT	US-22
FR-18	The system records question, answer, feedback, escalation and resolution events without storing message text in event data.	Must	Support Manager	Reconciliation test	US-23
FR-19	A manager dashboard shows volume, deflection rate, escalation rate, positive-feedback rate and top fallback topics with a date filter.	Must	Support Manager	UAT	US-24
FR-20	Conversation data is retained for 90 days by default (configurable), then deleted or anonymised automatically.	Must	Security Officer	Accelerated test	US-26
FR-21	An Admin can delete a customer's account and conversations on request; the action is logged.	Must	Security Officer	Test	US-26

ID	Requirement	Priority	Source	Verification	Stories
FR-22	KB changes and agent access to transcripts are written to an audit log that Admins can view.	Should	Support Manager	Test	US-08

Non-Functional Requirements

ID	Requirement	Priority	Source	Verification	Stories
NFR-01	Performance: p95 AI response time is 8 seconds or less with 50 concurrent sessions; error rate below 1%.	Must	Sponsor	Load test	US-27
NFR-02	Availability: 99.5% during pilot business hours (Mon-Fri 08:00-20:00); alerts on failure.	Must	Sponsor	Monitoring	US-29
NFR-03	Scalability: supports 50 concurrent chat sessions at pilot and can scale horizontally without redesign.	Should	CTO	Load test	US-27
NFR-04	Usability: a new agent completes open, reply and resolve after a 60-minute training; usability issues rated severity 1 are fixed before build.	Should	Support Manager	Usability test + training	US-03, US-30
NFR-05	Accessibility: customer chat meets WCAG 2.1 AA for keyboard navigation and colour contrast on key screens.	Should	Product Manager	Accessibility check	US-16
NFR-06	Compatibility: latest two versions of Chrome, Edge, Safari and Firefox; responsive down to 360px width.	Must	Product Manager	Cross-browser test	US-16
NFR-07	Maintainability: all code reviewed via pull request; automated tests run in CI; API documented with OpenAPI.	Should	CTO	CI + review	US-01
NFR-08	Observability: central logs, error alerts and metrics for latency and LLM spend.	Must	CTO	Inspection	US-29
NFR-09	Recovery: daily database backups; recovery point objective 24 hours, recovery time objective 4 hours (pilot).	Should	CTO	Restore test	US-02

Technical Requirements

ID	Requirement	Priority	Source	Verification	Stories
TR-01	Hosted on a single cloud region with separate dev, staging and production environments.	Must	CTO	Inspection	US-01
TR-02	CI/CD pipeline builds, tests and deploys to staging on merge; production release requires manual approval.	Must	CTO	Inspection	US-01, US-29
TR-03	A relational database (PostgreSQL) stores users, articles, conversations, escalations and events; vector search uses the pgvector extension.	Must	Backend Developer	Inspection	US-02, US-12
TR-04	The LLM and embedding models are called through an internal abstraction layer so a provider can be swapped without changing business logic.	Must	AI/ML Engineer	Code review	US-11, US-13
TR-05	REST API versioned and documented with OpenAPI; request/response contracts agreed before UI integration.	Should	Backend Developer	Inspection	US-13, US-16
TR-06	A transactional email provider delivers OTP codes and notifications with SPF/DKIM configured.	Must	Backend Developer	Test	US-06, US-22
TR-07	Secrets are held in a managed secret store; none are committed to source control.	Must	Security Officer	Scan + review	US-07

Security Requirements

ID	Requirement	Priority	Source	Verification	Stories
SR-01	All traffic uses TLS 1.2 or higher; data is encrypted at rest.	Must	Security Officer	Scan	US-01, US-02
SR-02	Passwords are hashed (bcrypt or argon2); sessions use secure, HTTP-only tokens.	Must	Security Officer	Code review + test	US-04
SR-03	Rate limiting is applied to authentication and chat endpoints.	Must	Security Officer	Test	US-07
SR-04	All inputs are validated and outputs encoded to prevent injection and cross-site scripting; at least 20 prompt-injection cases are tested.	Must	Security Officer	Security test	US-07
SR-05	No Critical or High vulnerability is open at go-live (dependency scan + security test).	Must	Security Officer	Scan + test	US-07
SR-06	Least privilege: each role can only reach the data and functions its permission matrix grants; every denial is logged.	Must	Security Officer	Access-control test	US-05

AI-Related Requirements

ID	Requirement	Priority	Source	Verification	Stories
AIR-01	Grounding: every AI answer is generated only from retrieved KB passages and cites at least one source.	Must	Support Manager	Golden-set evaluation	US-13
AIR-02	Accuracy: on a golden set of at least 100 questions, answer accuracy meets the Proposed Project Target of 85% or higher before go-live.	Must	Sponsor, Product Manager	Golden-set evaluation	US-15
AIR-03	Hallucination limit: unsupported answers are 5% or fewer across the golden and adversarial sets (Proposed Project Target).	Must	Sponsor, Security Officer	Golden-set evaluation	US-15, US-14
AIR-04	Safe fallback: at least 95% of unanswerable test questions trigger the fallback and escalation offer.	Must	Support Manager	Golden-set evaluation	US-14
AIR-05	Transparency: customers are told they are chatting with an AI assistant and can reach a human from every conversation.	Must	Product Manager, Legal	UAT	US-16, US-19
AIR-06	Provider data handling: written provider terms confirm no training on company data and minimal retention before any real customer data is sent.	Must	Security Officer, Legal	Document review	US-11, US-25
AIR-07	Cost control: per-conversation token cap and spend alerts at 70% and 90% of the 1,500 USD LLM budget.	Must	Project Manager, Finance	Monitoring	US-13, US-29
AIR-08	Traceability: model, prompt and retrieval-settings versions are stored with every answer.	Should	AI/ML Engineer	Inspection	US-02, US-13
AIR-09	Human oversight: during the pilot, the Support Manager reviews a sample of 20 AI conversations per week.	Should	Support Manager	Process check	US-24

Key definitions

Term	Definition used in this project
Deflection rate	Conversations with at least one AI answer that end without an escalation request, divided by all conversations with at least one AI answer.
Grounded answer	An answer whose statements are supported by the retrieved knowledge-base passages shown as sources.
Unsupported (hallucinated) answer	An answer containing a statement not supported by the retrieved passages, judged by a human reviewer on the golden set.
Golden set	At least 100 reviewed questions (including 20 unanswerable and 10 adversarial) with reference articles, used to measure answer quality every sprint.
Fallback	The assistant's response when confidence is below the threshold: it says it cannot answer and offers a human.

Traceability summary

61 requirements: 8 business, 22 functional, 9 non-functional, 7 technical, 6 security and 9 AI-related. Every functional, non-functional, technical, security and AI requirement maps to at least one of the 30 MVP stories; business requirements map to the functional requirements that realise them. US-28 (UAT) verifies all functional requirements.